CSC380 Homework 1 Problem 2: Data Analysis and Visualization

Overview This homework will familiarize you with the basic steps involved in reading, analyzing, and visualizing data. We will use the Starbucks Nutrition Dataset which itemizes most of the food and drink (12oz) options available at the Starbucks coffee chain. To simplify things we have processed the data for you into a JSON file distributed with the homework (filename: starbucks.json). We will be using the Pandas library to load and manipulate data. I briefly introduced all of the Pandas functionality that will need in class and additional links are provided inline below.

The problem has a maximum score of 70 points.

Installing Pandas To install any python library just type:

!pip3 install "library name"

Or, if you are using Anaconda then type:

!conda install "library name"

The cell below can be used to install Pandas. Or you can do it on the command line.

In [182…
```
# Uncomment and run the line below to install Pandas using pip
#!pip3 install pandas

# Uncomment and run the line below to install Pandas using Anaconda
#!conda install pandas
```

In [183…
```
import pandas as pd
```

Problem 1 : Basic Operations and Stats from the dataset

Download the data "starbucks.json" and load it to create a Pandas DataFrame.

What is a python DataFrame ? - https://www.geeksforgeeks.org/python-pandas-dataframe/

Hint : Check out the read_json function - https://www.w3schools.com/python/pandas/pandas_json.asp

In [184…
```
starbucks_df = pd.read_json('starbucks.json')
print(starbucks_df.to_string())
```

```
                    Beverage_category
Beverage Beverage_prep  Calories  Total Fat (g)  Trans Fat (g)  Saturated Fa
t (g)  Sodium (mg)  Total Carbohydrates (g)  Cholesterol (mg)  Dietary Fibre
(g)  Sugars (g)  Protein (g)  Vitamin A (fDV)  Vitamin C (fDV)  Calcium (fD
V)  Iron (fDV)  Caffeine (mg)
0                              Coffee
Brewed Coffee         Plain         5            0.1            0.0
0.0            0                        10                0
0           0          1.0             0.00             0.00           0.00
0.00            330
1             Classic Espresso Drinks
Caffè Latte   Nonfat Milk       130            0.3            0.2
0.0            5                       150               19
0          18         13.0             0.20             0.00           0.40
0.00            150
2             Classic Espresso Drinks
Caffè Latte       2% Milk       190            7.0            3.5
0.2           30                       170               19
0          17         12.0             0.20             0.02           0.40
0.00            150
3             Classic Espresso Drinks
Caffè Latte       Soymilk       150            5.0            0.5
0.0            0                       130               13
1           8         10.0             0.15             0.00           0.40
0.15            150
4             Classic Espresso Drinks                 Caffè Mocha (Without W
hipped Cream)   Nonfat Milk       220            2.5            1.5
0.0            5                       125               43
2          34         13.0             0.20             0.00           0.35
0.25            175
5             Classic Espresso Drinks                 Caffè Mocha (Without W
hipped Cream)       2% Milk       260            8.0            4.5
0.2           25                       140               42
2          34         13.0             0.15             0.02           0.35
0.25            175
6             Classic Espresso Drinks                 Caffè Mocha (Without W
hipped Cream)       Soymilk       230            7.0            2.0
0.0            0                       105               37
3          26         11.0             0.10             0.00           0.35
0.40            175
7             Classic Espresso Drinks            Vanilla Latte (Or Other Fl
avoured Latte)   Nonfat Milk       200            0.3            0.2
0.0            5                       140               37
0          35         12.0             0.20             0.00           0.35
0.00            150
8             Classic Espresso Drinks            Vanilla Latte (Or Other Fl
avoured Latte)       2% Milk       250            6.0            3.5
0.2           25                       150               37
0          35         12.0             0.20             0.02           0.35
0.00            150
9             Classic Espresso Drinks            Vanilla Latte (Or Other Fl
avoured Latte)       Soymilk       210            5.0            0.5
0.0            0                       120               31
1          26          9.0             0.10             0.00           0.35
0.15            150
10            Classic Espresso Drinks                                    C
```

```
aff Americano         Plain        15            0.0            0.0
0.0            0                       15                3
0           0          1.0             0.00             0.00           0.02
0.00            225
11            Classic Espresso Drinks
Cappuccino   Nonfat Milk        80            0.2            0.1
0.0            5                       90                12
0          10          8.0             0.15             0.00           0.25
0.00            150
12            Classic Espresso Drinks
Cappuccino       2% Milk       120            4.0            2.0
0.1           15                      100                12
0          10          8.0             0.10             0.00           0.25
0.00            150
13            Classic Espresso Drinks
Cappuccino       Soymilk       100            3.5            0.4
0.0            0                       80                 9
1           5          7.0             0.08             0.00           0.25
0.10            150
14            Classic Espresso Drinks                           Skinny Latte
(Any Flavour)   Nonfat Milk       120            0.3            0.2
0.0            5                      160                18
0          16         12.0             0.20             0.00           0.35
0.00            150
15          Signature Espresso Drinks                                  Car
amel Macchiato   Nonfat Milk       190            1.0            1.0
0.0           10                      130                35
0          32         11.0             0.20             0.00           0.35
0.00            150
16          Signature Espresso Drinks                                  Car
amel Macchiato       2% Milk       240            7.0            3.5
0.2           25                      150                34
0          32         10.0             0.15             0.02           0.30
0.00            150
17          Signature Espresso Drinks                                  Car
amel Macchiato       Soymilk       200            5.0            1.0
0.0            5                      115                29
1          24          9.0             0.10             0.00           0.35
0.15            150
18          Signature Espresso Drinks        White Chocolate Mocha (Without
Whipped Cream)   Nonfat Milk       350            6.0            4.5
0.0           10                      240                61
0          58         15.0             0.20             0.02           0.45
0.02            150
19          Signature Espresso Drinks        White Chocolate Mocha (Without
Whipped Cream)       2% Milk       400           11.0            7.0
0.2           25                      250                61
0          58         15.0             0.15             0.02           0.45
0.00            150
20          Signature Espresso Drinks        White Chocolate Mocha (Without
Whipped Cream)       Soymilk       370           10.0            5.0
0.0            0                      220                56
1          51         13.0             0.10             0.02           0.45
0.15            150
21          Signature Espresso Drinks              Hot Chocolate (Without
Whipped Cream)   Nonfat Milk       240            2.5            1.5
```

```
0.0            5                       140               48
2          41         14.0             0.20             0.00           0.40
0.25             25
22          Signature Espresso Drinks               Hot Chocolate (Without
Whipped Cream)       2% Milk       290            9.0            5.0
0.2            25                      160               47
2          41         14.0             0.20             0.02           0.40
0.25             25
23          Signature Espresso Drinks               Hot Chocolate (Without
Whipped Cream)       Soymilk       250            7.0            2.0
0.0            0                       125               41
3          32         12.0             0.15             0.00           0.40
0.40             25
24          Signature Espresso Drinks          Caramel Apple Spice (Without
Whipped Cream)         Plain       280            0.0            0.0
0.0            0                       20                70
0          65          0.0             0.00             0.00           0.00
0.00              0
25                  Tazo® Tea Drinks
Tazo® Tea          Plain         0            0.0            0.0
0.0            0                        0                 0
0           0          0.0             0.00             0.00           0.00
0.00             95
26                  Tazo® Tea Drinks                                  Tazo®
Chai Tea Latte   Nonfat Milk       210            0.2            0.1
0.0            5                       105               45
0          43          8.0             0.15             0.00           0.25
0.02             95
27                  Tazo® Tea Drinks                                  Tazo®
Chai Tea Latte       2% Milk       240            4.5            2.0
0.1           20                       115               45
0          42          8.0             0.10             0.00           0.25
0.02             95
28                  Tazo® Tea Drinks                                  Tazo®
Chai Tea Latte       Soymilk       220            3.5            0.4
0.0            0                        90               41
1          37          6.0             0.08             0.00           0.25
0.10             95
29                  Tazo® Tea Drinks                                 Tazo® G
reen Tea Latte   Nonfat Milk       290            0.5            0.2
0.0           10                       160               57
1          56         14.0             0.25             0.15           0.45
0.04             80
30                  Tazo® Tea Drinks                                 Tazo® G
reen Tea Latte       2% Milk       350            8.0            4.0
0.2           30                       180               57
1          55         14.0             0.20             0.15           0.45
0.04             80
31                  Tazo® Tea Drinks                                 Tazo® G
reen Tea Latte       Soymilk       310            6.0            1.0
0.0            0                       140               50
3          46         11.0             0.15             0.15           0.45
0.25             80
32                  Tazo® Tea Drinks                                 Tazo® Full-
Leaf Tea Latte   Nonfat Milk       150            0.2            0.1
0.0            5                        85               31
```

```
0          31          7.0             0.15            0.00           0.25
0.00            120
33                  Tazo® Tea Drinks                            Tazo® Full-
Leaf Tea Latte      2% Milk       190           4.0             2.0
0.1           15                        95                31
0          30          7.0             0.10            0.00           0.25
0.00            120
34                  Tazo® Tea Drinks                            Tazo® Full-
Leaf Tea Latte      Soymilk       170           3.5             0.4
0.0            0                        80                27
1          25          6.0             0.08            0.00           0.25
0.10            120
35                  Tazo® Tea Drinks     Tazo® Full-Leaf Red Tea Latte (Va
nilla Rooibos)   Nonfat Milk       150           0.2             0.1
0.0            5                        85                31
0          31          7.0             0.15            0.00           0.25
0.00              0
36                  Tazo® Tea Drinks     Tazo® Full-Leaf Red Tea Latte (Va
nilla Rooibos)      2% Milk       190           4.0             2.0
0.1           15                        95                31
0          30          7.0             0.10            0.00           0.25
0.00              0
37                  Tazo® Tea Drinks     Tazo® Full-Leaf Red Tea Latte (Va
nilla Rooibos)      Soymilk       170           3.5             0.4
0.0            0                        80                27
1          25          6.0             0.08            0.00           0.25
0.10              0
38              Shaken Iced Beverages             Iced Brewed Coffee (With
Classic Syrup)         Plain        90           0.1             0.0
0.0            0                         5                21
0          21          0.3             0.00            0.00           0.00
0.00            165
39              Shaken Iced Beverages       Iced Brewed Coffee (With Milk &
Classic Syrup)   Nonfat Milk       110           0.1             0.0
0.0            0                        30                24
0          24          2.0             0.04            0.00           0.08
0.02             90
40              Shaken Iced Beverages       Iced Brewed Coffee (With Milk &
Classic Syrup)      2% Milk       120           1.5             0.5
0.0            5                        35                24
0          24          2.0             0.04            0.00           0.08
0.00            125
41              Shaken Iced Beverages       Iced Brewed Coffee (With Milk &
Classic Syrup)      Soymilk       110           1.0             0.1
0.0            0                        30                23
0          22          2.0             0.02            0.00           0.08
0.00            125
42              Shaken Iced Beverages          Shaken Iced Tazo® Tea (With
Classic Syrup)         Plain        80           0.0             0.0
0.0            0                         0                21
0          21          0.0             0.00            0.00           0.00
0.00            100
43              Shaken Iced Beverages  Shaken Iced Tazo® Tea Lemonade (With
Classic Syrup)         Plain       130           0.0             0.0
0.0            0                         0                33
0          33          0.1             0.00            0.15           0.00
```

```
0.00            100
44                          Smoothies                           Banana Choc
olate Smoothie   Nonfat Milk       280            2.5            1.5
0.0             5                       150                53
7           34          20.0             0.10            0.15           0.20
0.00            100
45                          Smoothies                           Banana Choc
olate Smoothie       2% Milk       300            5.0            2.5
0.1           15                    160                53
7           34          20.0             0.08            0.15           0.20
0.20            15
46                          Smoothies                           Banana Choc
olate Smoothie       Soymilk       290            4.5            1.5
0.0             5                       150                51
7           31          19.0             0.06            0.15           0.20
0.20            15
47                          Smoothies                        Orange Mango B
anana Smoothie   Nonfat Milk       260            1.0            0.3
0.0             5                       120                54
6           37          16.0             0.50            0.80           0.10
0.30            15
48                          Smoothies                        Orange Mango B
anana Smoothie       2% Milk       270            1.5            0.5
0.0             5                       125                53
6           37          16.0             0.50            0.80           0.10
0.06             0
49                          Smoothies                        Orange Mango B
anana Smoothie       Soymilk       270            1.5            0.4
0.0             5                       120                53
6           36          15.0             0.50            0.80           0.10
0.06             0
50                          Smoothies                          Strawberry B
anana Smoothie   Nonfat Milk       290            1.0            0.3
0.0             5                       125                58
7           41          16.0             0.04            1.00           0.10
0.08             0
51                          Smoothies                          Strawberry B
anana Smoothie       2% Milk       290            2.0            1.0
0.0             5                       125                58
7           41          16.0             0.04            1.00           0.10
0.08             0
52                          Smoothies                          Strawberry B
anana Smoothie       Soymilk       290            2.0            0.4
0.0             5                       120                58
8           40          16.0             0.02            1.00           0.10
0.08             0
53        Frappuccino® Blended Coffee
Coffee   Nonfat Milk       220            0.1            0.1
0.0             0                       210                51
0           50           4.0             0.06            0.00           0.10
0.04            70
54        Frappuccino® Blended Coffee
Coffee    Whole Milk       240            3.0            2.0
0.1            10                       220                50
0           50           3.0             0.04            0.00           0.10
0.00            95
```

```
55        Frappuccino® Blended Coffee
Coffee       Soymilk       220            1.5            0.2
0.0            0                       210               49
0          47          3.0             0.04            0.00           0.10
0.00             95
56        Frappuccino® Blended Coffee                        Mocha (Without
Whipped Cream)   Nonfat Milk       260            1.0            0.5
0.0            0                       220               61
1          58          4.0             0.06            0.00           0.10
0.08            110
57        Frappuccino® Blended Coffee                        Mocha (Without
Whipped Cream)    Whole Milk       290            4.0            2.5
0.1           10                       220               61
1          58          4.0             0.04            0.00           0.10
0.08            110
58        Frappuccino® Blended Coffee                        Mocha (Without
Whipped Cream)       Soymilk       260            2.5            0.5
0.0            0                       220               59
1          55          4.0             0.04            0.00           0.10
0.15            110
59        Frappuccino® Blended Coffee                      Caramel (Without
Whipped Cream)   Nonfat Milk       280            0.1            0.1
0.0            0                       220               60
0          59          4.0             0.06            0.00           0.10
0.00            100
60        Frappuccino® Blended Coffee                      Caramel (Without
Whipped Cream)    Whole Milk       280            3.5            2.0
0.1           10                       220               60
0          59          3.0             0.04            0.00           0.10
0.00            100
61        Frappuccino® Blended Coffee                      Caramel (Without
Whipped Cream)       Soymilk       260            1.5            0.2
0.0            0                       220               58
0          56          3.0             0.04            0.00           0.10
0.06            100
62        Frappuccino® Blended Coffee                       Java Chip (Without
Whipped Cream)   Nonfat Milk       310            4.0            3.0
0.0            0                       250               67
2          62          5.0             0.06            0.00           0.10
0.25            110
63        Frappuccino® Blended Coffee                       Java Chip (Without
Whipped Cream)    Whole Milk       340            7.0            5.0
0.1           10                       250               67
2          62          5.0             0.04            0.00           0.10
0.25            110
64        Frappuccino® Blended Coffee                       Java Chip (Without
Whipped Cream)       Soymilk       310            6.0            3.5
0.0            0                       240               65
2          60          4.0             0.04            0.00           0.10
0.30            110
65  Frappuccino® Light Blended Coffee
Coffee   Nonfat Milk       120              0.1            0.1
0.0            0                       210               26
0          26          3.0             0.06            0.00           0.10
0.00             95
66  Frappuccino® Light Blended Coffee
```

```
Mocha   Nonfat Milk       150            1.0            0.5                0
.0            0                      200               33
1          30          4.0             0.06            0.00           0.10
0.08             95
67  Frappuccino® Light Blended Coffee
Caramel   Nonfat Milk       150            0.1            0.1
0.0            0                      200               33
0          32          3.0             0.06            0.00           0.10
0.00             90
68  Frappuccino® Light Blended Coffee
Java Chip   Nonfat Milk       220            4.0            3.0
0.0            0                      240               43
2          39          5.0             0.06            0.00           0.10
0.25            105
69         Frappuccino® Blended Crme         Strawberries & Crme (Without Wh
ipped Cream)   Nonfat Milk       230            0.2            0.1
0.0            0                      190               53
0          52          4.0             0.08            0.06           0.15
0.04             0
70         Frappuccino® Blended Crme         Strawberries & Crme (Without Wh
ipped Cream)    Whole Milk       260            4.0            2.0
0.1           10                      190               53
0          52          4.0             0.06            0.06           0.15
0.04             0
71         Frappuccino® Blended Crme         Strawberries & Crme (Without Wh
ipped Cream)       Soymilk       240            2.0            0.2
0.0            0                      180               51
1          49          3.0             0.04            0.06           0.15
0.08             0
72         Frappuccino® Blended Crme                Vanilla Bean (Without W
hipped Cream)   Nonfat Milk       240            0.1            0.1
0.0            5                      230               56
0          55          5.0             0.08            0.00           0.15
0.00             0
```

Printing the entire dataframe looks cumbersome. How can we look at the first and last **two** rows of a dataframe?

Check out .head() and .tail() - https://www.tutorialspoint.com/python_pandas/python_pandas_basic_functionality.htm

What are the first two and last two rows on the dataframe?

In [185…

```
print("First 2 rows of the dataframe:\n", starbucks_df.head(2))
print()
print("Last 2 rows of the dataframe:\n", starbucks_df.tail(2))
```

```
First 2 rows of the dataframe:
         Beverage_category       Beverage Beverage_prep  Calories  \
0                   Coffee  Brewed Coffee         Plain         5
1  Classic Espresso Drinks    Caff Latte   Nonfat Milk       130

   Total Fat (g)  Trans Fat (g)  Saturated Fat (g)  Sodium (mg)  \
0            0.1            0.0                0.0            0
1            0.3            0.2                0.0            5

   Total Carbohydrates (g)  Cholesterol (mg)  Dietary Fibre (g)  Sugars (g)
\
0                       10                 0                  0           0
1                      150                19                  0          18

   Protein (g)  Vitamin A (fDV)  Vitamin C (fDV)  Calcium (fDV)  Iron (fDV)
\
0          1.0              0.0              0.0            0.0         0.0
1         13.0              0.2              0.0            0.4         0.0

   Caffeine (mg)
0            330
1            150

Last 2 rows of the dataframe:
              Beverage_category                                      Beverag
e  \
71  Frappuccino® Blended Crme  Strawberries & Crme (Without Whipped Cream)
72  Frappuccino® Blended Crme           Vanilla Bean (Without Whipped Cream)

   Beverage_prep  Calories  Total Fat (g)  Trans Fat (g)  Saturated Fat (g)
\
71       Soymilk       240            2.0            0.2                0.0
72   Nonfat Milk       240            0.1            0.1                0.0

    Sodium (mg)  Total Carbohydrates (g)  Cholesterol (mg)  Dietary Fibre
(g)  \
71            0                      180                51
1
72            5                      230                56
0

    Sugars (g)  Protein (g)  Vitamin A (fDV)  Vitamin C (fDV)  Calcium (fDV)
\
71          49          3.0             0.04             0.06           0.15
72          55          5.0             0.08             0.00           0.15

    Iron (fDV)  Caffeine (mg)
71        0.08              0
72        0.00              0
```

How can we access just a column of a dataset in pandas? https://cmdlinetips.com/2020/04/3-ways-to-select-one-or-more-columns-with-pandas/.

It is okay if while printing you only see first and last few element and dots in between; this is Python's way of summarizing the output.

Print the column 'Beverage_prep'

In [186…
```
print(starbucks_df.Beverage_prep.to_string())
```

```
0           Plain
1     Nonfat Milk
2         2% Milk
3         Soymilk
4     Nonfat Milk
5         2% Milk
6         Soymilk
7     Nonfat Milk
8         2% Milk
9         Soymilk
10          Plain
11    Nonfat Milk
12        2% Milk
13        Soymilk
14    Nonfat Milk
15    Nonfat Milk
16        2% Milk
17        Soymilk
18    Nonfat Milk
19        2% Milk
20        Soymilk
21    Nonfat Milk
22        2% Milk
23        Soymilk
24          Plain
25          Plain
26    Nonfat Milk
27        2% Milk
28        Soymilk
29    Nonfat Milk
30        2% Milk
31        Soymilk
32    Nonfat Milk
33        2% Milk
34        Soymilk
35    Nonfat Milk
36        2% Milk
37        Soymilk
38          Plain
39    Nonfat Milk
40        2% Milk
41        Soymilk
42          Plain
43          Plain
44    Nonfat Milk
45        2% Milk
46        Soymilk
47    Nonfat Milk
48        2% Milk
49        Soymilk
50    Nonfat Milk
51        2% Milk
52        Soymilk
53    Nonfat Milk
54     Whole Milk
55        Soymilk
```

```
56    Nonfat Milk
57     Whole Milk
58        Soymilk
59    Nonfat Milk
60     Whole Milk
61        Soymilk
62    Nonfat Milk
63     Whole Milk
64        Soymilk
65    Nonfat Milk
66    Nonfat Milk
67    Nonfat Milk
68    Nonfat Milk
69    Nonfat Milk
70     Whole Milk
71        Soymilk
72    Nonfat Milk
```

One goal of data science is to use data in order to answer questions. This is done in an automated way, without us having to manually go through the data. Let's try answering some simple questions about Starbucks' menu items.

a. On an average, how much caffeine does a starbucks drink have?

Hint: Checkout the math functions of a pandas dataframe.

https://erikrood.com/Python_References/pandas_column_average_median_final.html

In [187…
```
print(starbucks_df['Caffeine (mg)'].mean())
```

```
95.75342465753425
```

b. What is the *typical* (median) amount of caffeine in a starbucks drink?

In [188…
```
print(starbucks_df['Caffeine (mg)'].median())
```

```
100.0
```

c. What is the maximum amount of caffeine you can find at starbucks in its drinks?

In [189…
```
print(starbucks_df['Caffeine (mg)'].max())
```

```
330
```

d. What is the least amount of caffeine you can find at starbucks in its drinks?

In [190…
```
print(starbucks_df['Caffeine (mg)'].min())
```

0

Problem 2 : Pie Chart

Let's explore the dataset we have a bit more further

a. What are the different type of Drinks (ie Beverage Category)that Starbucks has? How much of each?

Hint - Checkout pandas value_counts() function.

In [191…
```
beverage_category_counts = starbucks_df["Beverage_category"].value_counts()
print(beverage_category_counts)
```

```
Beverage_category
Classic Espresso Drinks              14
Tazo® Tea Drinks                     13
Frappuccino® Blended Coffee          12
Signature Espresso Drinks            10
Smoothies                             9
Shaken Iced Beverages                 6
Frappuccino® Light Blended Coffee     4
Frappuccino® Blended Crme            4
Coffee                                1
Name: count, dtype: int64
```

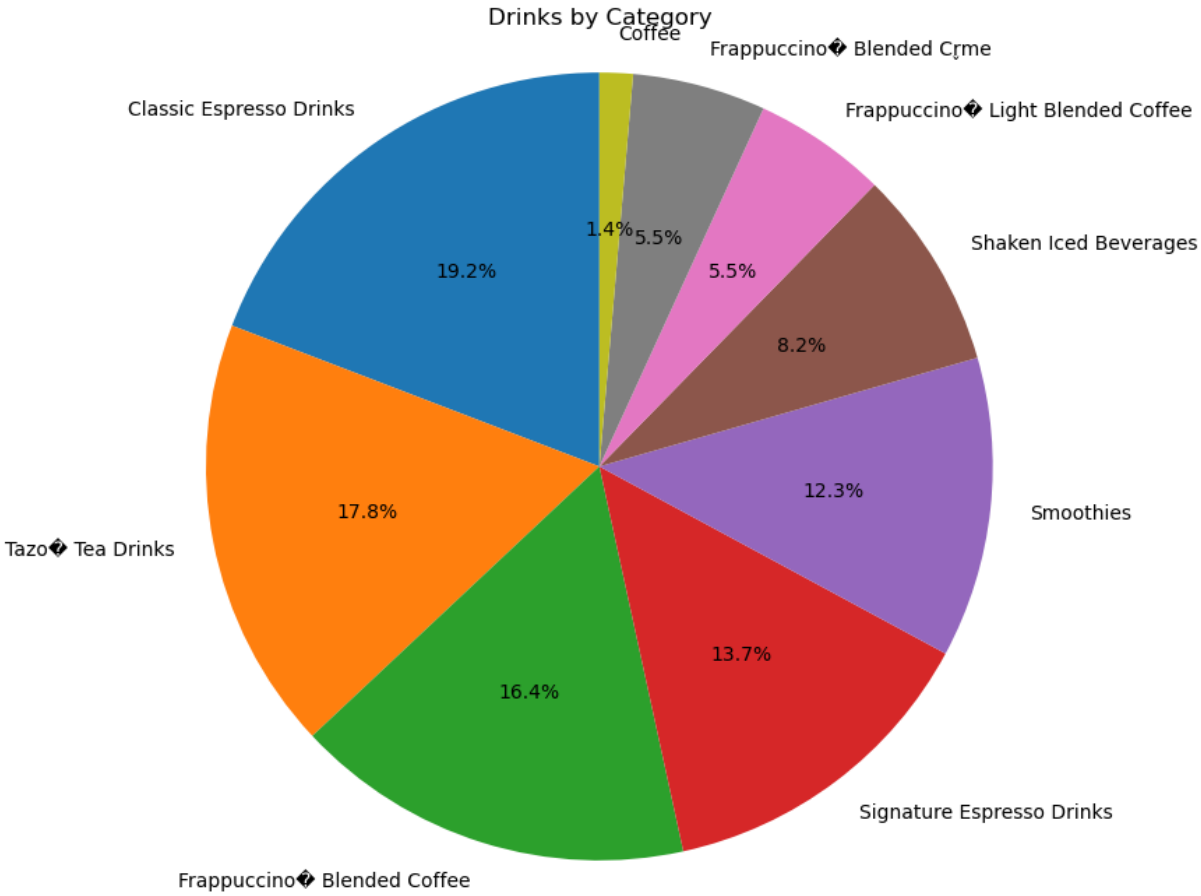
Let's make these more appealing. Plot these as a pie chart

In [192…
```
import matplotlib.pyplot as plt

beverage_category_counts = starbucks_df["Beverage_category"].value_counts()
labels = beverage_category_counts.index
sizes = beverage_category_counts.values

plt.figure(figsize=(8,8))
plt.pie(sizes, labels=labels, autopct="%1.1f%%", startangle=90)
plt.title("Drinks by Category")
plt.axis("equal")

plt.show()
```



Problem 3 : Bar Chart

Suppose you have a very calorie conscious friend. But they really like to get the drinks at Starbucks. As a budding Data Scientist, you want to help them out.

a. What is the drink with the least amount of calories at Starbucks?

Hint : Check this out ==> https://www.interviewqs.com/ddi-code-snippets/rows-cols-python

In [193…
```
min_cals = starbucks_df["Calories"].min()
drinks_with_least_cals = starbucks_df.loc[starbucks_df["Calories"] == min_ca

print(drinks_with_least_cals.to_string(index=False))
```

```
Tazo® Tea
```

But they are quickly bored of this drink. I mean, it's only natural.

So, let's recommend them a beverage category instead.

First let's find on an average how much calories do each beverage category have?

Hint - Checkout groupby function. The first example in this page is what we are trying to do. https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.groupby.html

In [194…
```
avg_calories = starbucks_df.groupby("Beverage_category")["Calories"].mean()
print(avg_calories)
```

```
Beverage_category
Classic Espresso Drinks               162.500000
Coffee                                  5.000000
Frappuccino® Blended Coffee           272.500000
Frappuccino® Blended Crme             242.500000
Frappuccino® Light Blended Coffee     160.000000
Shaken Iced Beverages                 106.666667
Signature Espresso Drinks             281.000000
Smoothies                             282.222222
Tazo® Tea Drinks                      203.076923
Name: Calories, dtype: float64
```

b. Plot a bar Graph

Let's make this visually appealing by plotting a bar graph, where the height of the bar plot is average amount of calories.

Hint : Check this out -> https://benalexkeen.com/bar-charts-in-matplotlib/

In [195…
```
import matplotlib.pyplot as plt

means = starbucks_df.groupby("Beverage_category")["Calories"].mean()

x = means.index.tolist()
y = means.values.tolist()

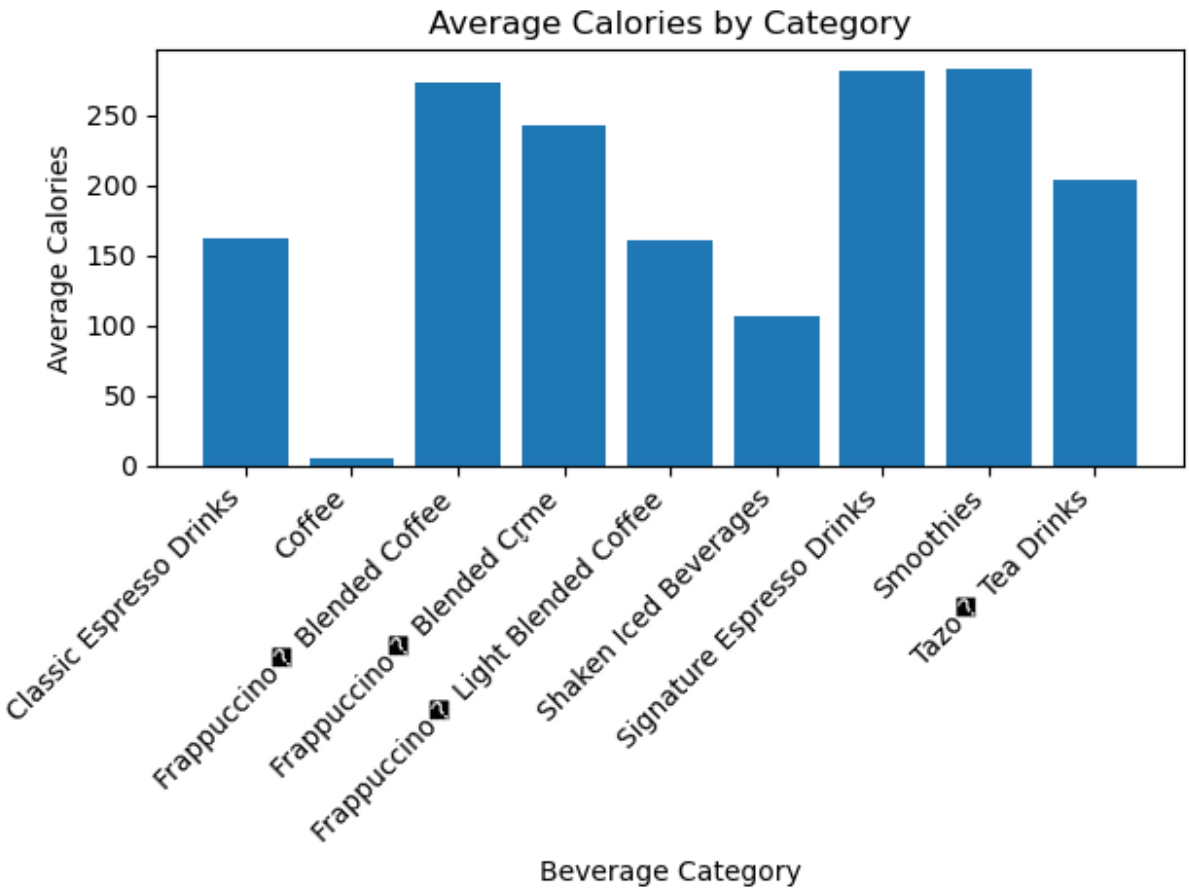
x_pos = [i for i, _ in enumerate(x)]

plt.bar(x_pos, y)
plt.ylabel("Average Calories")
plt.xlabel("Beverage Category")
plt.title("Average Calories by Category")

plt.xticks(x_pos, x)

plt.setp(plt.gca().get_xticklabels(), rotation=45, horizontalalignment="righ

plt.tight_layout()
plt.show()
```



By looking at the graph, which beverage category has the least average calories?

In [196…
```
print('Coffee')
```

```
Coffee
```

Let's keep looking

By looking at the graph, which beverage category has the second least average calories?

In [197…
```
print('Shaken Iced Beverages')
```

```
Shaken Iced Beverages
```

This gives us some idea of how many calories to expect in each beverage category. But we know from our previous classes that taking just the mean is not a good representation of how the values are spread. In this case, while the average is useful, we need to know how it is spread across various drinks within a beverage category.

What is the standard deviation of calories within each beverage categories?

Standard deviation is a concept we haven't yet covered in class (as of 1/22). But its idea

is simple: similar to the range of the data we saw, it measures the "spread" of the data.

Hint: try replacing the aggregate function after groupby() from mean() to std().

In [198…
```
std_calories = starbucks_df.groupby("Beverage_category")["Calories"].std()
print(std_calories)
```

```
Beverage_category
Classic Espresso Drinks              71.649521
Coffee                                     NaN
Frappuccino® Blended Coffee          36.711405
Frappuccino® Blended Crme           12.583057
Frappuccino® Light Blended Coffee    42.426407
Shaken Iced Beverages                18.618987
Signature Espresso Drinks            71.561939
Smoothies                            13.017083
Tazo® Tea Drinks                     87.405627
Name: Calories, dtype: float64
```

If you get a nan for Coffee in the above cell, just add .fillna(0) at the end. To read more about fillna(0) - https://pandas.pydata.org/pandas-docs/stable/reference/api/pandas.DataFrame.fillna.html

Else skip the below cell

In [199…
```
std_calories = starbucks_df.groupby("Beverage_category")["Calories"].std().f
print(std_calories)
```

```
Beverage_category
Classic Espresso Drinks              71.649521
Coffee                                0.000000
Frappuccino® Blended Coffee          36.711405
Frappuccino® Blended Crme           12.583057
Frappuccino® Light Blended Coffee    42.426407
Shaken Iced Beverages                18.618987
Signature Espresso Drinks            71.561939
Smoothies                            13.017083
Tazo® Tea Drinks                     87.405627
Name: Calories, dtype: float64
```

Now Let's incorporate this info into the bar chart as well. We want a bar chart where there is 1 bar for each beverage category, the height is average calories, and error bars representing +/- sample standard deviation. Hint: go back to https://benalexkeen.com/bar-charts-in-matplotlib/

In [200…
```
import matplotlib.pyplot as plt

means = starbucks_df.groupby("Beverage_category")["Calories"].mean()
stds = starbucks_df.groupby("Beverage_category")["Calories"].std().fillna(0)

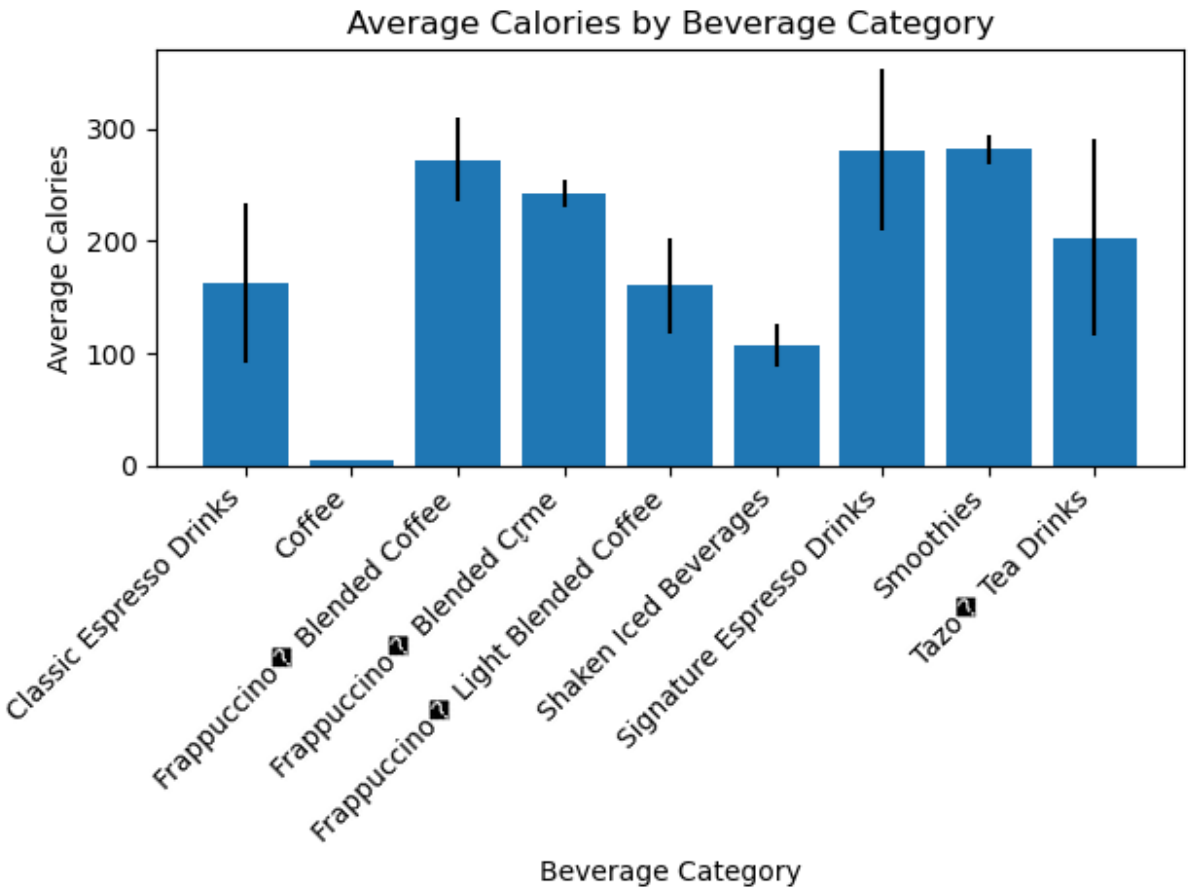
x = means.index.tolist()
y = means.values.tolist()
```

```
variance = stds.values.tolist()

x_pos = [i for i, _ in enumerate(x)]
plt.bar(x_pos, y, yerr=variance)
plt.ylabel("Average Calories")
plt.xlabel("Beverage Category")
plt.title("Average Calories by Beverage Category")

plt.xticks(x_pos, x)
plt.setp(plt.gca().get_xticklabels(), rotation=45, horizontalalignment="righ

plt.tight_layout()
plt.show()
```



Look how easy it is to understand that many numbers when visualised well!

Awesome work so far!!

Problem 4 : Scatter plot

Now another friend of yours, who absolutely loves Caffeine came to you for a recommendation.They want to know what are the top drinks with the most Caffeine in Starbucks. They would like to know how much sugar each of them may have too, since they would like to reduce that.They don't like numbers much,so we want to present this to them in a attractive way. Let's start by sorting the DataFrame based on Caffeine

Hint: https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.sort_values.html

In [201…
```
sorted_list = starbucks_df.sort_values(by="Caffeine (mg)", ascending=False)
sorted_list.head()
```

Out[201…

	Beverage_category	Beverage	Beverage_prep	Calories	Total Fat (g)	Trans Fat (g)	Saturated Fat (g)
0	Coffee	Brewed Coffee	Plain	5	0.1	0.0	0.0
10	Classic Espresso Drinks	Caffè Americano	Plain	15	0.0	0.0	0.0
4	Classic Espresso Drinks	Caffè Mocha (Without Whipped Cream)	Nonfat Milk	220	2.5	1.5	0.0
5	Classic Espresso Drinks	Caffè Mocha (Without Whipped Cream)	2% Milk	260	8.0	4.5	0.2
6	Classic Espresso Drinks	Caffè Mocha (Without Whipped Cream)	Soymilk	230	7.0	2.0	0.0

What are the top 10 **drinks** with the most caffeine in them?

Hint : Remember.head() from earlier. Use that.

In [202…
```
top10_Caf_Drink = sorted_list.head(10)
top10_Caf_Drink
```

Out[202...

	Beverage_category	Beverage	Beverage_prep	Calories	Total Fat (g)	Trans Fat (g)	Saturated Fat (g)
0	Coffee	Brewed Coffee	Plain	5	0.1	0.0	0.0
10	Classic Espresso Drinks	Caffè Americano	Plain	15	0.0	0.0	0.0
4	Classic Espresso Drinks	Caffè Mocha (Without Whipped Cream)	Nonfat Milk	220	2.5	1.5	0.0
5	Classic Espresso Drinks	Caffè Mocha (Without Whipped Cream)	2% Milk	260	8.0	4.5	0.2
6	Classic Espresso Drinks	Caffè Mocha (Without Whipped Cream)	Soymilk	230	7.0	2.0	0.0
38	Shaken Iced Beverages	Iced Brewed Coffee (With Classic Syrup)	Plain	90	0.1	0.0	0.0
14	Classic Espresso Drinks	Skinny Latte (Any Flavour)	Nonfat Milk	120	0.3	0.2	0.0
1	Classic Espresso Drinks	Caffè Latte	Nonfat Milk	130	0.3	0.2	0.0
20	Signature Espresso Drinks	White Chocolate Mocha (Without Whipped Cream)	Soymilk	370	10.0	5.0	0.0
19	Signature Espresso Drinks	White Chocolate Mocha (Without Whipped Cream)	2% Milk	400	11.0	7.0	0.2

We don't really care about the other nutritions at this point. Let's just print what is needed.

In [203…
```
top10_Caf_Drink[['Beverage','Sugars (g)','Caffeine (mg)']]
```

Out[203…

	Beverage	Sugars (g)	Caffeine (mg)
0	Brewed Coffee	0	330
10	Caff̧ Americano	0	225
4	Caff̧ Mocha (Without Whipped Cream)	34	175
5	Caff̧ Mocha (Without Whipped Cream)	34	175
6	Caff̧ Mocha (Without Whipped Cream)	26	175
38	Iced Brewed Coffee (With Classic Syrup)	21	165
14	Skinny Latte (Any Flavour)	16	150
1	Caff̧ Latte	18	150
20	White Chocolate Mocha (Without Whipped Cream)	51	150
19	White Chocolate Mocha (Without Whipped Cream)	58	150

Oops, why does the same drink keep repeating but with different calories and caffeine? Give yourself a minute before reading the next line for the answer.

Yes, they are prepared differently. Let's add that too, since it is relevent information

In [204…
```
top10_Caf_Drink[['Beverage','Beverage_prep','Sugars (g)','Caffeine (mg)']]
```

Out[204…

	Beverage	Beverage_prep	Sugars (g)	Caffeine (mg)
0	Brewed Coffee	Plain	0	330
10	Caff̧ Americano	Plain	0	225
4	Caff̧ Mocha (Without Whipped Cream)	Nonfat Milk	34	175
5	Caff̧ Mocha (Without Whipped Cream)	2% Milk	34	175
6	Caff̧ Mocha (Without Whipped Cream)	Soymilk	26	175
38	Iced Brewed Coffee (With Classic Syrup)	Plain	21	165
14	Skinny Latte (Any Flavour)	Nonfat Milk	16	150
1	Caff̧ Latte	Nonfat Milk	18	150
20	White Chocolate Mocha (Without Whipped Cream)	Soymilk	51	150
19	White Chocolate Mocha (Without Whipped Cream)	2% Milk	58	150

Now that we have the beverages with the prep, sugar and caffeine,we need to show this to our friend. Let's plot them as a need scatter plot. Caffeine on x, Sugars on y.

In [205…

```
x = top10_Caf_Drink["Caffeine (mg)"].to_list()
y = top10_Caf_Drink["Sugars (g)"].to_list()

beverages = top10_Caf_Drink['Beverage'].to_list()
beverage_prep = top10_Caf_Drink['Beverage_prep'].to_list()
labels = [ str(beverages[i]) + ' with' + str(beverage_prep[i]) for i in rang

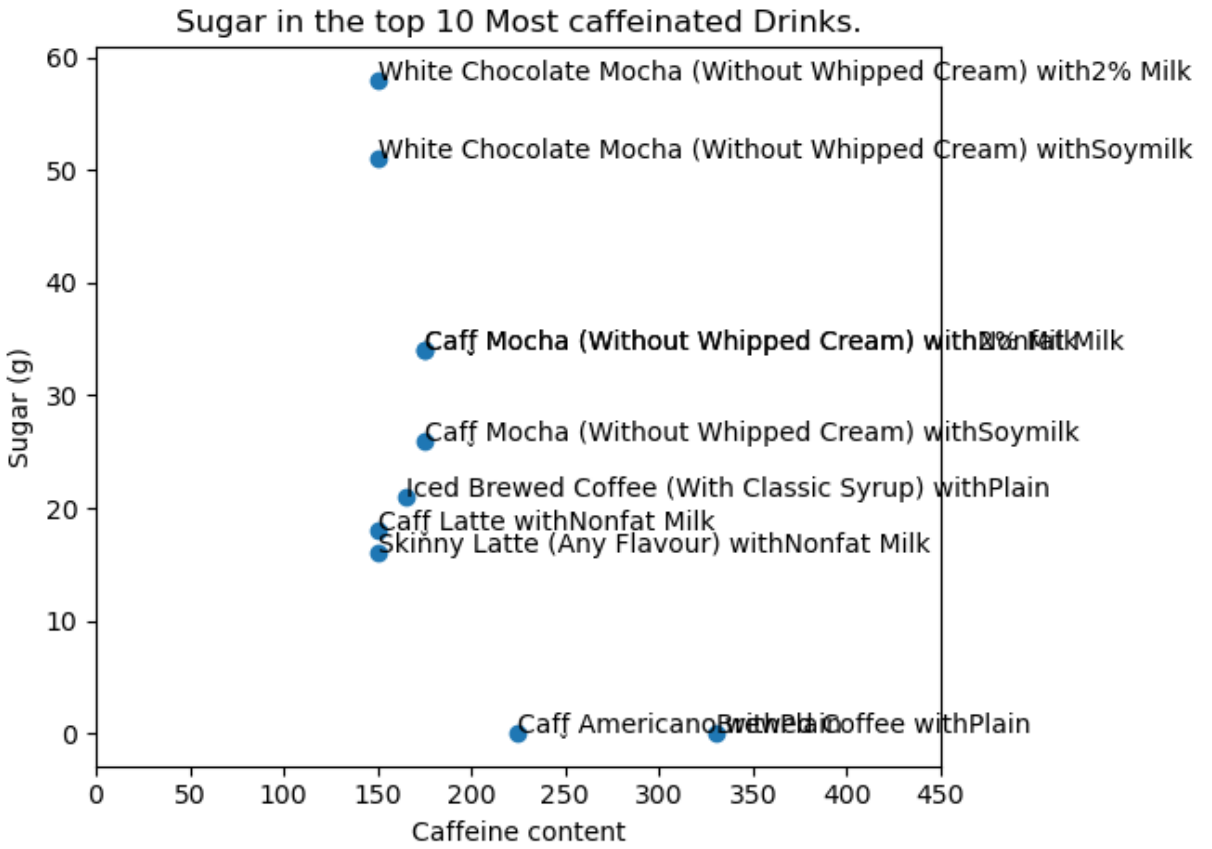
plt.scatter(x, y)

plt.xlabel("Caffeine content")
plt.ylabel("Sugar (g)")
plt.title("Sugar in the top 10 Most caffeinated Drinks.")

axes = plt.gca()
axes.set_xlim([0,450])

for i, txt in enumerate(labels):
    plt.annotate(txt, (x[i], y[i]))

plt.tight_layout()
```



Nice work!!

Conclusion

In this assignment, we were able to download a dataset, load it as a pandas dataframe,

explore the dataset with basic statistical functions and visulaise many specifc examples to answer relevent queries from the topic.

Congragulations!!

In [ ]: